UNIT 8

CYBERSECURITY AND INTERNET ETHICS

Learning Objectives

1. Identify common online threats, including viruses, malware, phishing, and hacking.
2. Understand safe browsing practices and password security measures.
3. Recognize the importance of ethical behaviour online, including avoiding plagiarism and respecting privacy.

The internet has become an essential part of our daily lives. It helps us learn, communicate, and explore the world. However, as we use the internet, we must also protect ourselves from online threats and behave responsibly. Cybersecurity refers to the practices and technologies designed to safeguard computers, networks, and data from harmful attacks. Internet ethics, on the other hand, guides us on how to use the internet respectfully and responsibly.

In this chapter, we will explore common online threats, ways to stay safe online, and how to use the internet ethically.

Common Online Threats

The internet is full of opportunities, but it also has risks. Understanding common online threats can help us protect ourselves.

1.Viruses

Viruses are malicious programs designed to harm your computer or steal your data. They can spread through infected files, emails, or software.

How to Avoid Viruses:

- Do not open emails or files from unknown sources.
- Use antivirus software and keep it updated.

2. Malware

Malware refers to harmful software, including viruses, spyware, and ransomware. It can damage your device or steal personal information.

How to Avoid Malware:

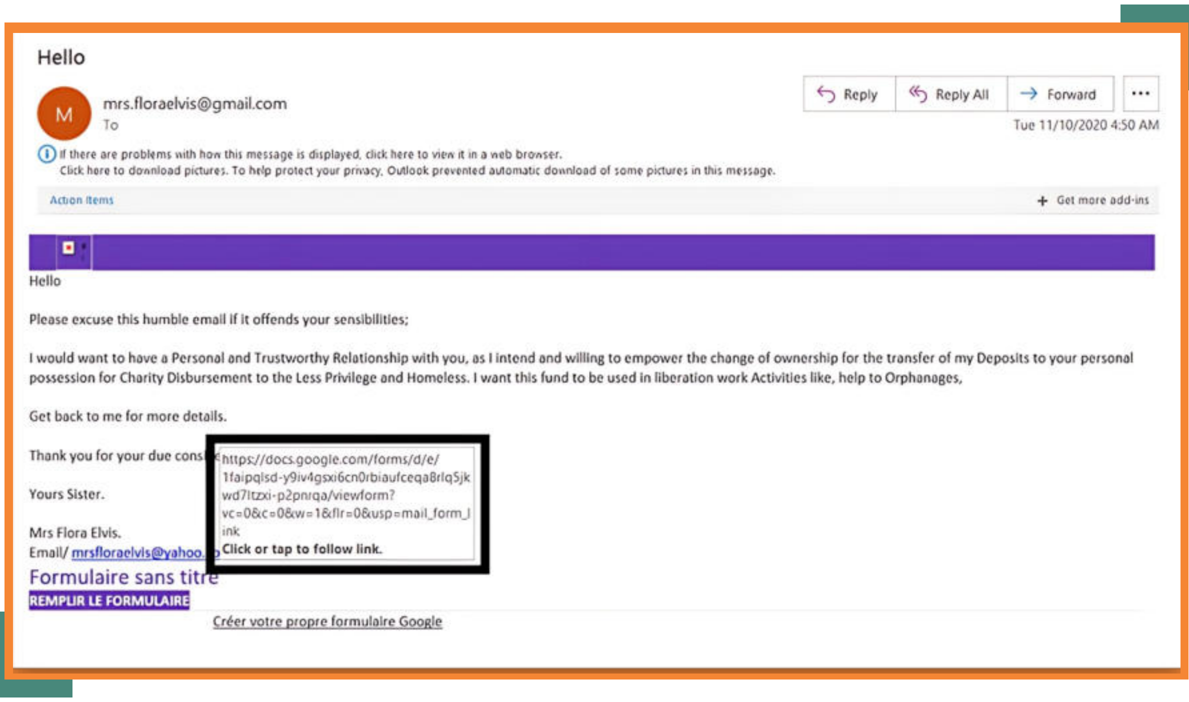
- Avoid downloading files from untrusted websites.
- Install and regularly update anti-malware programs.

3. Phishing

Phishing is a scam where cybercriminals trick you into sharing personal information, such as passwords or credit card details, by pretending to be trustworthy organizations.

How to Avoid Phishing:

- Do not click on suspicious links in emails or messages.
- Verify the sender's email address before providing sensitive information.



4. Hacking

Hacking involves gaining unauthorized access to your computer or accounts to steal information or cause harm.

How to Avoid Hacking:

- Use strong and unique passwords for each account.
- Enable two-factor authentication for additional security.

Safe Browsing Practices and Password Security

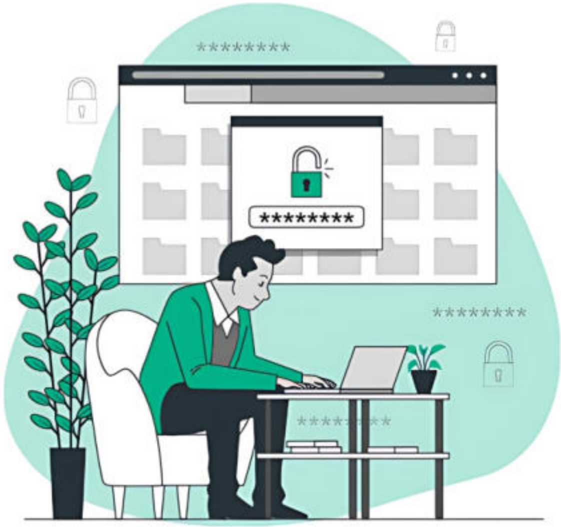
Staying safe online requires cautious behaviour and secure practices.

Safe Browsing Practices

- Avoid visiting unknown or untrusted websites.
- Look for the padlock symbol in the browser's address bar to ensure the website is secure.
- Be careful when sharing personal information online.

Password Security

- Use strong passwords that include a mix of letters, numbers, and symbols.
- Avoid using the same password for multiple accounts.
- Change your passwords regularly.



How To Choose
A Strong Password

Ethical Use of the Internet

Using the internet responsibly ensures that it remains a safe and respectful space for everyone.

- **Avoiding Plagiarism**

Plagiarism is copying someone else's work or ideas without giving them credit. It is unethical and can have serious consequences.

How to Avoid Plagiarism:

- Always cite your sources when using someone else's work.
- Use quotation marks for direct quotes.
- Paraphrase information and give proper credit.

- **Respecting Privacy**

Respecting others' privacy is essential for ethical internet use. Avoid sharing or accessing private information without permission.

How to Respect Privacy:

- Do not share others' photos, videos, or personal details without their consent.
- Be mindful of your own privacy settings on social media platforms.

The internet is a powerful tool, but it comes with responsibilities. Understanding and avoiding online threats, practising safe browsing, and behaving ethically ensures a safe and respectful online experience. By following these guidelines, we can protect ourselves and contribute to a positive digital environment.

EXERCISES

01: Choose the correct answer.

1.Which of the following is an example of malware?

a) Microsoft Word

b) Antivirus software

c) Spyware

d) Email

2.What symbol indicates a secure website in the browser's address bar?

a) Star

b) Padlock

c) Warning sign

d) Key

3.Phishing attempts often occur through:

a) Phone calls

b) Fake emails

c) Social media posts

d) Online shopping

4.A strong password should include:

a) Only letters

b) Only numbers

c) A mix of letters, numbers, and symbols

d) Repeated words

5.What is the purpose of antivirus software?

a) To edit photos

b) To protect against viruses

c) To browse the internet

d) To store data

6.Hacking refers to:

a) Fixing a computer

b) Unauthorized access to a system

c) Writing code

d) Installing software

7.What does HTTPS stand for?

a) Hypertext Transfer Protocol Secure

b) Hyperlink Transfer Protocol Secure

c) Hypertext Technology Secure

d) Hyper Transfer Protocol System

8.Plagiarism means:

a) Using someone's work without credit

b) Respecting privacy online

c) Writing original content

d) Downloading software

9.Which of these is a safe browsing practice?

a) Sharing passwords

b) Visiting unknown websites

c) Checking for the padlock icon

d) Ignoring antivirus updates

10.Respecting privacy online involves:

a) Sharing others' photos without consent

b) Avoiding social media

c) Protecting personal information

d) Using weak passwords

02: Fill in the blanks.

1. ______ is a malicious program that can harm your computer or steal data.

2. Phishing scams often trick people into sharing ______ information.

3. A ______ password includes letters, numbers, and symbols.

4. Avoid downloading files from ______ websites to stay safe online.

5. Respecting others' ______ is an essential part of internet ethics.

03: Select True or False for each statement.

	TRUE	FALSE
Malware can include viruses, spyware, and ransomware.		
Using the same password for multiple accounts is a good practice.		
HTTPS is more secure than HTTP.		
Plagiarism is acceptable if you only use a small part of someone's work.		
Hacking always refers to fixing computer problems.		
Antivirus software helps protect your computer from threats.		

04: Answer the following questions briefly.

1. What is phishing, and how can you protect yourself from it?

2. Name two ways to ensure your passwords are secure.

3. Why is respecting privacy important in internet ethics?

4. What does a padlock symbol in a browser indicate?

5. Explain the term "malware."

05: Provide detailed answers for the following questions.

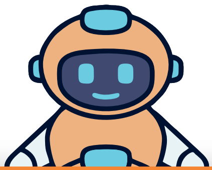
1. Describe three common online threats and how to protect yourself from them.

2. Why is cybersecurity important in today's digital world? Provide examples.

3. Explain the importance of password security and provide tips for creating strong passwords.

4. Discuss the role of internet ethics in maintaining a safe and respectful online environment.

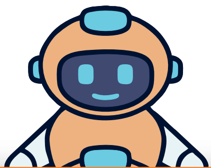
Lab Activity 1: Creating Strong Passwords



Objective: Teach students how to create strong and secure passwords.

Instructions: Provide examples of weak and strong passwords. Ask students to create strong passwords for hypothetical accounts and explain why they are secure.

Lab Activity 2: Identifying Phishing Attempts



Objective: Help students recognize phishing emails and websites.

Instructions: Show examples of real and fake emails. Students will identify suspicious elements and explain their reasoning.